DOME: A Domain-Orchestrated Multi-GPU Optical Network for Rack-Scale Systems

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***Abstract*—Modern data centers increasingly use multi-GPU systems for AI and high-performance computing, where growing data transfer demands lead to high energy consumption and performance bottlenecks in electrical networks. Optical interconnects offer compelling advantages to address these challenges, including high bandwidth, distance-independent latency, and better energy efficiency. This paper presents DOME, a rack-scale optical interconnection network that connects multiple GPUs using high-radix optical switches and extends optical interfaces into GPU packages close to memories and multiprocessors, forming distinct in-GPU and GPU-to-GPU network domains. To efficiently manage the paths across switches and domains, we develop a multi-switch arbitration scheme and a time-slotted path reservation scheme that quickly identifies the earliest time when the path is available in all network domains, reducing unnecessary reservation retries. Evaluations reveal that DOME achieves 14% speedup while maintaining comparable energy consumption compared to the state-of-the-art preemptive chain feedback control scheme.**

***Index Terms*—optical interconnection networks, GPUs, path reservation, rack-scale systems**

I. Introduction

Large-scale scientific computations and large language models have dramatically increased computational demands, driving the adoption of tens of hundreds of GPUs that fill an entire rack in data center [1] [2]. Unlike CPUs, GPUs feature massively parallel execution units requiring frequent access to large memory chunks via the interconnection network. This creates unprecedented bandwidth intensity where network power consumption scales prohibitively with system size [3].

Current multi-GPU systems rely on copper-based electrical links for communication, but face GPU-specific constraints: Their extreme bandwidth demands [4] [5] require numerous parallel wires [6] [7], straining GPU package pin limits and causing thermal/packaging challenges [8]. Unlike CPUs, GPUs' dense integration exacerbates signal integrity issues over meter-long distances in-rack [9] [10], restricting per-link bandwidth and elevating energy consumption.

Optical interconnects offer superior performance characteristics, providing long-distance communication with an order of magnitude greater bandwidth at comparable energy consumption to electrical counterpart [9] [10]. Wavelength division multiplexing (WDM) enables multiple data streams per fiber [9], with recent demonstrations achieving 400 Gb/s and 10× energy efficiency gains [11]. These advantages uniquely address GPU bottlenecks by enabling high-bandwidth communication. Emerging 2.5D/3D co-packaged optics directly integrate optical interfaces near GPU dies [12], alleviating pin constraints while providing high intra/inter-GPU bandwidth. Industry implementations include Ayar Labs' TeraPHY [13] [14] and Intel's Optical Compute Interconnect [15], alongside academic advances [16] [17] [18]. These optical interconnects, together with microring (MR)-based optical switching elements, can be connected and form a larger optical switch to connect the GPUs.

Deploying optical interconnects and optical switches in rack-scale multi-GPU systems introduces complex system-level challenges that can prevent realizing their full bandwidth potential. While each optical link offers high bandwidth, achieving high bandwidth throughout the entire optical network requires coordinating multiple optical switches and multiple network domains - both within GPU packages and between GPUs. The optical network's circuit-switched nature, where complete paths must be configured before data transfer, fundamentally differs from electrical networks' store-and-forward operation. Consequently, path configuration becomes a coordination problem across multiple optical switches and domains that, if not properly managed, can significantly degrade network resource utilization and system performance [19] [20].

While extensive research exists on CPU-oriented networks, approaches like FlexiShare [21] and Corona [22] employ complex routing and arbitration schemes optimized for diverse, latency-sensitive CPU traffic patterns. On-chip and off-chip co-design efforts [19] [20] coordinate optical paths across domains but prioritize fairness through request-level scheduling—suited for CPU workloads with short messages and real-time constraints, which are not optimal for GPUs' bandwidth-intensive bulk transfers. Some works target at optical interconnects for a single GPU [23] [24], but they neglect GPU-to-GPU communication, while off-chip-only approaches [25] do not address the GPU package pin constraints and potentially create bandwidth mismatch between electrical on-chip and optical off-chip network domains. SiP-ML [26] and SiPAC [27] co-design deep neural network (DNN) communication algorithms with optical switches, but our work focuses on bandwidth utilization-oriented optical network management over diverse common GPU applications.

To this end, this paper presents **DOME**, a **D**omain-**O**rchestrated **M**ulti-GPU optical n**E**twork for rack-scale multi-GPU systems. Hundreds of GPUs communicate via multiple low-loss high-radix optical switches [28], with each GPU featuring a co-packaged optical die bridging in-GPU and GPU-to-GPU domains.

DOME addresses GPU-specific bandwidth and multi-domain coordination challenges through two control schemes: (1) adaptive multi-switch arbitration that dynamically allocates optical switches based on transfer size and utilization, enabling parallel transfers; (2) a time-slotted control scheme where network controllers exchange future slot availability across domains to reserve paths in advance, ensuring paths are configured only when time slots are available across all domains, thus eliminating unnecessary retries. It is worth noting that DOME operates exclusively at the network layer, thus it is compatible with existing GPU racks without requiring modifications to upper transport protocol layers or application stacks.

Our contributions are as follows:

1) We propose DOME, a fully optical interconnection network architecture for rack-scale multi-GPU systems using high-radix

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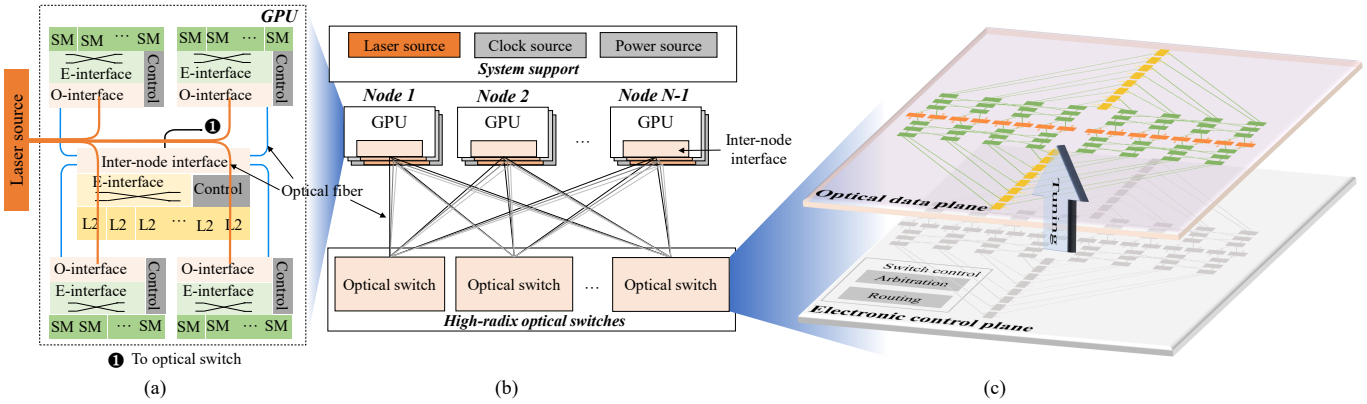


Fig. 1: Overview of the proposed multi-GPU system architecture

optical switches. The inter-node interface bridging the in-GPU and GPU-to-GPU network domains is detailed.

2) A dynamic multi-switch arbitration scheme and coordinated multi-domain control scheme are presented.
3) We quantitatively evaluate DOME in both network-scale and full system-scale, and compare performance and energy against handshake [29], PCF [20], and common electrical counterpart [30]. Implementation cost is also analyzed.

The rest of the paper is organized as follows. Section II outlines the DOME architecture. Section III details the multi-switch and multi-domain control scheme implemented in electronic control plane. Experimental setup and evaluation are depicted in Section IV and Section V. Finally, Section VI concludes the paper.

II. DOME Architecture

A. System Overview

Fig. 1 shows our multi-GPU system with nodes typically containing 4 GPUs per board. GPUs interconnect via centralized high-radix optical switches [28], enabling rack-scale deployment. Switch controllers and optical dies are separate co-packaged chiplets. Each GPU also features a chiplet-based architecture with hundreds of compute units (with private L1 caches) and shared memory partitions (called L2 caches), similar to industry designs [31], [32]. The compute unit chiplet connects to the memory partition chiplet via optical links for enhanced bandwidth.

Each GPU connects to the optical switch chips through the optical inter-node interface and optical fibers. Such co-packaged optical interface provides higher bandwidth density per pin than electrical alternatives, reducing pin constraints and package costs [8], [10].

B. Packaging

The chiplet-based GPU co-packages an optical inter-node interface die via 3D stacking (Fig. 2). The optical die connects to electronic dies (compute units/L2) through TSVs, integrating optical components including waveguides, MR-based switching elements, modulators, and photodetectors. This contrasts with monolithic integration, thus enables independent manufacturing/testing while mitigating thermal issues. Adjacent optical dies connect via optical fiber, as demonstrated in industry prototypes [13], [33]. Electrical signals pass through TSVs to the optical die for modulation, routing through waveguides and switching elements before photodetectors convert them back to electrical signals at the destination die.

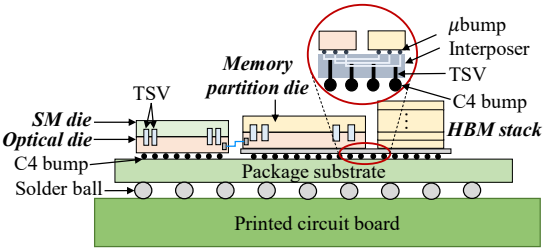


Fig. 2: Cross-section view of a chiplet-based GPU co-packaged with optical dies

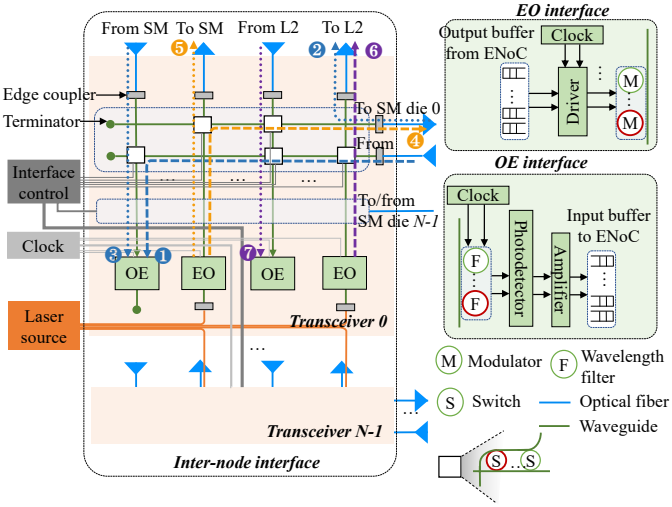


Fig. 3: Co-packaged optical inter-node interface bridging intra- and inter-GPU network domains. Data paths include: requests to ❶ local and ❷ remote L2, or ❸ local GPU; replies to ❹ local and ❺ remote GPU, ❻ large memory-to-memory data transfers, and ❼ memory copies to local memory partitions.

C. Inter-Node Interface

The inter-node interface bridges networks between chiplets within each GPU (intra-GPU, Fig. 1a) and between GPUs (inter-GPU, Fig. 1b), enabling direct end-to-end optical connectivity across both network domains. As shown in Fig. 3, it employs pairs of Electro-Optical (EO) and Optical-Electrical (OE) converters. Each EO/OE pair time-multiplexes electrical signals into optical wavelengths via

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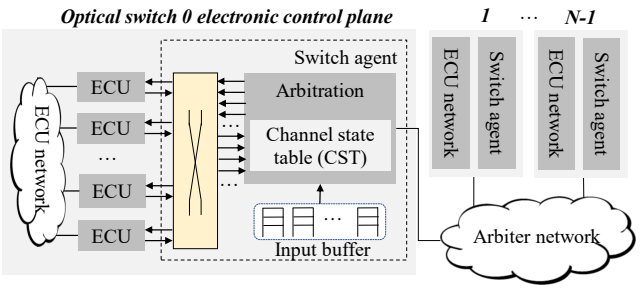


Fig. 4: Optical switch control plane enhanced with multi-switch arbitration

Algorithm 1 Multi-switch arbitration scheme

Require: IB: Input buffer, $SST[S][R][N]$: Switch status table
Ensure: Allocated paths A
1: $A \leftarrow \emptyset$; $N_a \leftarrow \sum_{s,i,j} SST[s][i][j]$
2: $F \leftarrow \min(N_a/\text{SIZE}(IB), 1)$ {Scaling factor}
3: **while** IB not empty **do**
4: $T \leftarrow \text{POP}(IB)$
5: $P_r \leftarrow \max(1, \lfloor \lceil T.\text{size}/BW_{\text{path}} \rceil \times F \rfloor)$
6: $L \leftarrow \emptyset$; $P_a \leftarrow 0$ {Path list and counter}
7: **for** s, i, j where $SST[s][i][j] = 1$ **and** $P_a < P_r$ **do**
8: $SST[s][i][j] \leftarrow 0$; $P_a \leftarrow P_a + 1$
9: $L.\text{APPEND}((s, i, j))$
10: **end for**
11: **if** $P_a > 0$ **then**
12: $A[T] \leftarrow L$
13: **else**
14: $\text{PUSH}(IB, T)$
15: **end if**
16: **end while**
17: **return** A

MR arrays. To prevent congestion between latency-sensitive short request/reply/control messages and bandwidth-sensitive large data transfers (e.g., RDMA operations), the interface implements dedicated separate paths for different traffic types as illustrated in Fig. 3.

III. Multi-Domain Optical Path Control

A. Multi-Switch Arbitration

1) **Adaptive Scaling:** To balance resource allocation under varying load, DOME computes a scaling factor comparing total available optical channels to pending transfers (Line 2).
2) **Path Requirement Estimation:** For each transfer, the arbiter calculates a baseline number of optical paths required to service the transfer, ensuring sufficient bandwidth (Line 5).
3) **Switch Allocation:** Given the scaling factor, the arbiter allocates physical channels across switches while guaranteeing at least one path per transfer (Line 6).

Multi-switch arbitration is implemented in the electronic control unit (ECU) to coordinate path allocation across distributed optical switches, as shown in Fig. 4. This scheme dynamically scales optical switch resource allocation based on transfer size and system-wide load, enabling parallel transfers while preventing resource monopolization. The algorithm operates in three steps:

The algorithm has time complexity $O(S \cdot R \cdot N + M \cdot S \cdot R \cdot N)$, where S is switches, R is rows per switch, N is channels per row, and M is pending transfers.

B. Chain Request-Feedback Control

Path configuration in our rack-scale multi-GPU system spans three network domains: source GPU's inter-node interface, optical switches, and destination GPU's inter-node interface. The traditional handshake approach requires multiple retries to find out common available time across domains, leading to wasted link bandwidth. DOME overcomes this through chain-based request-feedback control using synchronized time slots.

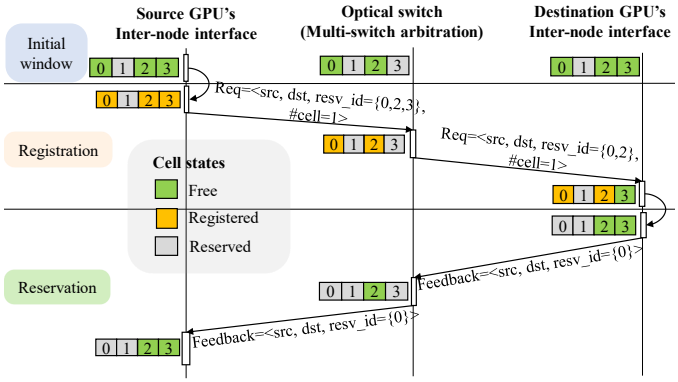


Fig. 5: Example of the chain request and feedback control

The workflow is illustrated in Fig. 5. Time is divided into fixed-length slots, each tracked by a *cell* storing availability state, path details, and a synchronized ID across domains. Matching cell IDs ensure aligned time references. Unlike traditional handshakes that lock paths immediately, DOME's reservation window lets GPUs reserve future slots in advance without retries. Before selected for reservation, the time that the request stays in the request queue is defined as **queuing time**. **Preemption time** measures the delay between reservation and actual path setup, while **transmission time** covers the optical data transfer duration.

DOME operates in two phases: *registration* and *reservation*. During registration, available slots propagate forward through domains while filtering occupied slots. In the example shown in Fig. 5, source GPU registers free slots 0/2/3, switches filter busy slot 3, and destination GPU selects the earliest free slot 0. In reservation, a feedback vector propagates backward to confirm the earliest common slot (e.g., slot 0), transitioning its state to "reserved" while releasing others. This forms a V-shaped control chain. Null feedback indicates failure, triggering retries. **Registration time** and **reservation time** denote phase durations, while **control time** aggregates message transmission delays.

Time slot length must exceed control overhead. As shown in Fig. 7, one V-shaped chain operation requires ∼40 ns to complete. To accommodate this control overhead and maintain proper cell synchronization, the time slot length must exceed this minimum threshold. We present a sensitivity analysis regarding time slot length in Section V-A.

IV. Methodology

In this section, we evaluate DOME against three other configurations (see Table I) under synthetic traces for network-level characteristics. Then, we analyze the full system using HPC workloads from Rodinia3.0 [34] and three hotspot kernels from transformer model training [35].

A. Optical Interconnect Model

The power and timing characteristics of optical interconnects are analyzed by OEIL [37]. It accounts for laser power, EO/OE conversion, MR thermal tuning, and attenuation derived from cumulative optical device loss. The laser power is configured to compensate

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TABLE I: Comparison of network configurations

Scheme	Description*
Electrical	Baseline electrical switching and electrical in-GPU networks with optical links between GPUs and the switch chip (aligned with NVLink [30]); requires intermediate EO/OE conversions
Handshake	Configure the path per domain once available, but forced to release the path once configuration in subsequent domains fails due to blocking [29]
PCF	Coordinated control with partial time slot reservation tailored for hybrid message size (common in CPU workloads); ongoing large data transfer may be interrupted and rescheduled for awaiting small ones to avoid starvation [20]
DOME	Coordinated control with full time slot reservation only; tailored for uniform, bulk messages from GPUDirect RDMA operations [36].

* Except for the electrical configuration, all other schemes employ optical switching and fully optical multi-domain connectivity.

TABLE II: Optical Interconnect Parameters

Component	Parameter / Value
Laser	EO power conversion efficiency: 30%
Device loss from data plane	Waveguide propagation: 0.5 dB/cm, waveguide bending: 0.05 dB/90°, edge coupling: 2 dB, MR insertion: 1 dB, MR passing: 0.06 dB
Photodetector	-20 dBm, 1.1 mW
MR	Tuning: 0.65 mW, static: 0.12 mW, dynamic power: 0.12 mW/Gbps

for the attenuation while maintain detectable signal strength at the receiver. Table II lists the key parameters used in the power model [37].

B. Multi-GPU System Simulation

We modifed Accel-Sim [38] and AccelWatch [39] to simulate the performance and power of the chiplet-based GPU architecture. The latency and energy consumption of the electrical and optical interconnect are analyzed by OEIL [37]. Table III details the multi-GPU system parameters.

TABLE III: Multi-GPU System Parameters

Parameter	Value
Compute subsystem	
System scale	128 GPUs total, 4 GPUs per node
SM	4 SM dies, 32 SMs per SM die, 32 warps per SM, round-robin warp scheduling, @1 GHz
Memory hierarchy	
L1 cache	32 KB per SM, 128 B lines, 4 ways
L2 cache	4 MB per chiplet (8 slices, 16-way, 256-set)
Global memory	4 GB per stack×4 stacks, 180 GB/s per stack, 100 ns, FR-FCFS scheduler
Electrical network	
Link	Inter-die: 16 B flit, 42 GB/s, 2 ns, 2 pJ/b; inter-GPU: 4 lanes × 50 Gb/s per lane, 4.21 pJ/b [37]; packet size (request/reply/memory copy/control): 1/8/8/1 flits
Switch	8 128-radix butterfly switches
Optical network	
Inter-node interface	8 transceivers per GPU
Link	Reply and memory copy network: 8 links, 50 Gbps×8 WDM channels per link; request and control network: 1 link, 50 Gbps×1 WDM channel per link
Switch	8 128-radix optical switches

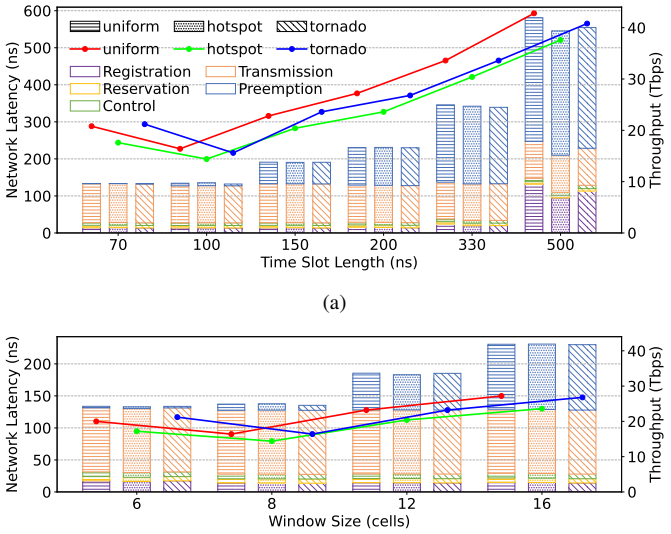


Fig. 6: Sensitivity analysis on (a) time slot length and (b) reservation window size.

V. Evaluation

A. Parameter Sensitivity Analysis

We analyze two key scheduling parameters under high network load: time slot length and reservation window size. As shown in Fig. 6, both parameters affect network latency and throughput.

Time slot length creates a tradeoff between path configuration latency and network throughput (Fig. 6a). Shorter slots enable more precise scheduling and reduce preemption time, but at lower link utilization due to reduced fraction of time dedicated to data transmission. However, longer slots allow for extended transmission planning, improving overall throughput despite increased preemption time. Based on these results, we select 200 ns as the desired time slot length.

The reservation window size analysis (Fig. 6b) shows that a 12-cell window achieves comparable throughput to 16 cells while reducing both latency and control overhead. The total latency comprises queuing time, reservation time, preemption delay, and optical transmission delay. The rest of the evaluations will follow these two values.

B. Network-Level Performance Analysis

1) Latency and Throughput: We evaluate message latency for GPU-to-GPU memory transfers under varying injection rates, focusing on uniform traffic patterns (Fig. 7). Under light network load (injection rate of 10 Tbps), the chain request-feedback operation is around 40 ns. While latency increases with injection rates across all configurations, DOME reduces latency by 9.8%, 26.7%, and 21.3% compared to electrical, handshake, and PCF schemes.

The handshake and PCF schemes exhibit higher control overhead than electrical networks due to their path reservation mechanisms. The handshake approach reserves paths domain-by-domain when available but releases pre-reserved paths upon failure, leading to significant queuing delays from repeated reservation attempts. In contrast, both PCF and DOME maintain minimal queuing delays. DOME achieves 45% lower registration latency than PCF by eliminating complex time management operations like partial cell reservation across network domains.

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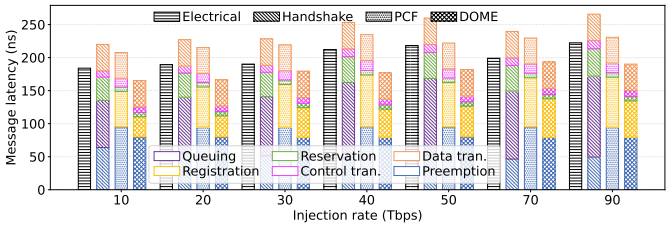


Fig. 7: Average message latency under uniform traffic pattern with respect to varying injection rates

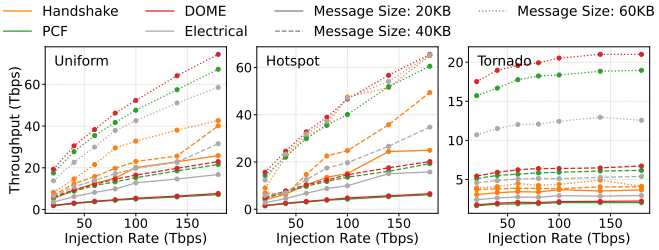


Fig. 8: Network throughput under varying traffic patterns and message size

Fig. 8 shows aggregate network throughput across different traffic patterns. Uniform traffic yields the highest throughput, while tornado traffic shows the lowest saturation throughput due to path conflicts. Under uniform and hotspot patterns, PCF and DOME may not match the throughput of electrical or handshake schemes. However, larger message sizes improve throughput efficiency as the longer transmission time better amortizes optical path setup overhead.

2) Path Setup Efficiency: We analyze multi-domain path reservation efficiency by measuring the distribution of reservation retries (Fig. 9). The handshake scheme exhibits poor reservation success rates, with over 60% of reservations requiring multiple retry attempts. PCF achieves marginally higher reservation success rates than DOME, likely due to its fine-grained partial time slot tracking that reduces scheduling gaps and accommodates more requests within the reservation window.

C. Power Consumption

For our 128-GPU rack deployment, we conduct a detailed power analysis at both the component and system levels, with the results presented in Fig. 10. At the component level, our 400 Gbps optical interconnect (comprising 8 WDM channels at 50 Gbps each) consumes 346.7 mW per end-to-end path, with 76.4% of the power attributed to MR thermal tuning. At the system level, the entire network consumes 500 W, with 74.4% dedicated to optical switches. This demonstrates significant power efficiency gains over electrical

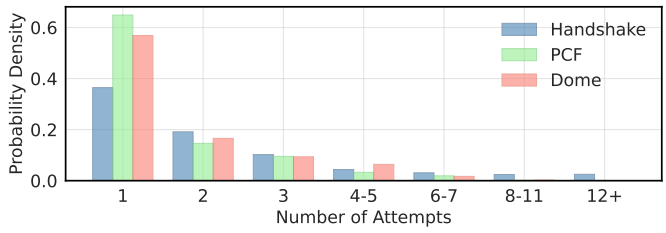
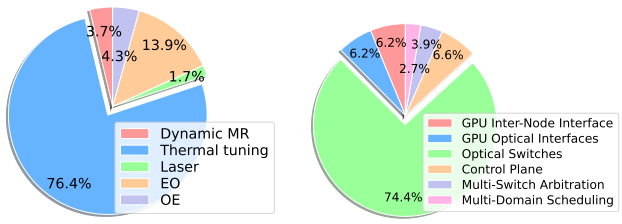


Fig. 9: Histogram of the number of retries in path reservation across domains



(a) (b)

Fig. 10: Power breakdown on optical network categorized by (a) optical components and (b) network organization

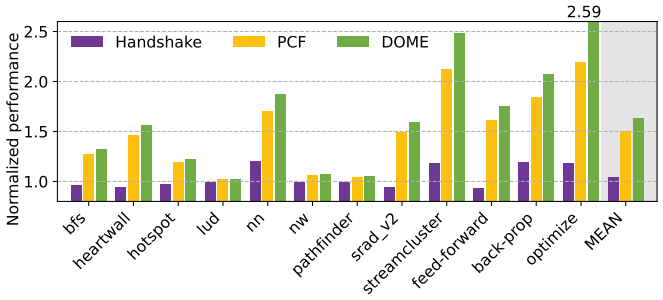


Fig. 11: System performance normalized to electrical switching configuration

switches [30], primarily due to fewer optical I/O ports and the elimination of intermediate EO/OE conversions. Additionally, the control plane accounts for 13.2% of the total network power.

D. Full-System Evaluation

1) System Performance: Fig. 11 shows system performance normalized to the electrical configuration. DOME achieves speedups of 1.64×, 1.6×, and 1.14× compared to electrical, handshake, and PCF schemes, respectively. The handshake scheme generally performs no better than the electrical baseline due to high reservation control overhead. Applications with massive bulk data transfers, particularly *nn*, *streamcluster*, and LLM workloads, benefit most from optical transmission.

To understand how optical networking impacts system performance, we analyze GPU-to-GPU memory access latency across domains (Fig. 12). These transfers, predominantly handled through bulk memory access and GPUDirect RDMA operations between GPU memories, show latency patterns inversely proportional to system performance. DOME reduces memory transfer latency by 7%–30% compared to other schemes. The three LLM kernels demonstrate the most significant improvement, achieving up to 75% latency reduction due to their large transfer sizes.

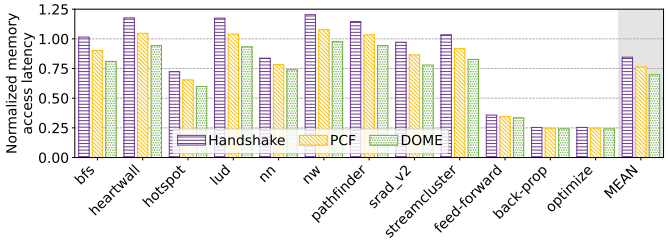


Fig. 12: GPU-to-GPU average memory access latency

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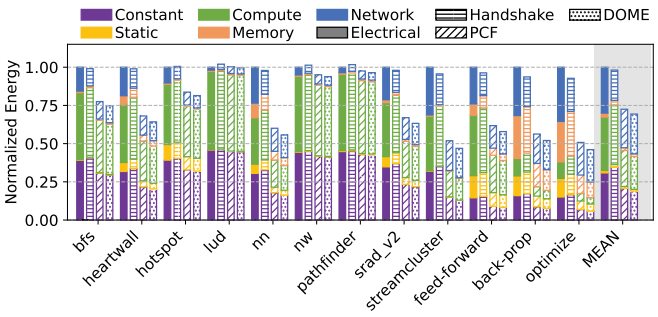


Fig. 13: System energy breakdown

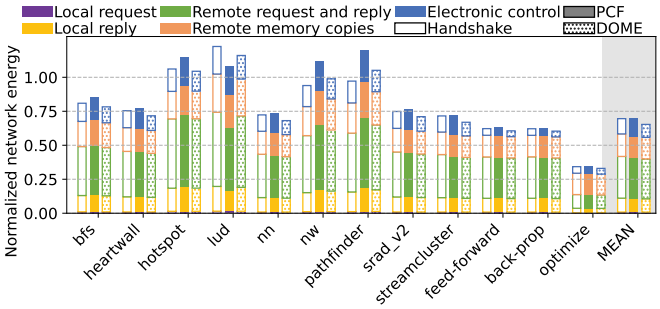


Fig. 14: Network energy breakdown

2) System Energy: Fig. 13 presents system energy breakdown normalized to the electrical baseline configuration. System power comprises constant, static, and dynamic components, with dynamic power primarily from compute, memory, and network subsystems. DOME reduces system power by 31%, 27%, and 3% compared to electrical, handshake, and PCF schemes, respectively. While power dissipation remains constant across schemes for compute and memory subsystems, total energy consumption decreases due to reduced execution time. Applications with intensive long message transfers, such as *nn*, *streamcluster*, and LLM workloads, show the highest energy reduction benefits from high-bandwidth optical networks.

Network energy maintains a similar proportion of total system energy across all schemes. The network energy consists of optical data plane and electronic control plane components. Network energy analysis across different traffic patterns (Fig. 14) shows comparable results among traffic types, with electronic control accounting for 14%–20% of total energy. This consistency stems from the shared optical data plane design across schemes. However, for applications with short message sizes (*hotspot*, *lud*, *nw*, and *pathfinder*), optical networks consume more energy than electrical networks due to higher electronic control overhead.

3) Energy-Delay Product: Fig. 15 shows the energy-delay product (EDP) normalized to the electrical network configuration. All three multi-domain optical networks achieve significantly lower EDP (38%–61%) compared to traditional electrical switches, demonstrating the energy efficiency of fully optical interconnects in multi-GPU systems. Through optimized optical path control, DOME further reduces EDP by 24% and 5% compared to handshake and PCF schemes, respectively.

E. Optical Die Implementation Cost

1) Inter-node Interface: We analyze the implementation cost through detailed hardware resource requirements and area efficiency

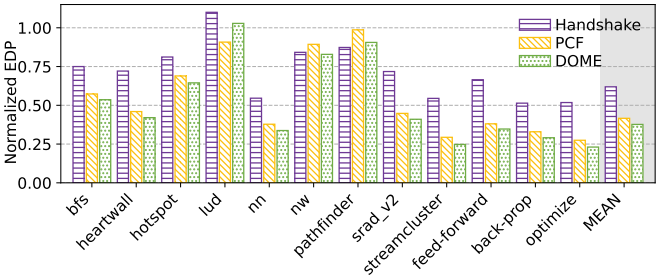


Fig. 15: Normalized energy-delay product under different multi-domain control schemes

metrics. As shown in Table IV, with 10 μm MR diameter and 2 μm waveguide pitch specifications, our design achieves high area efficiency: the SM-side and L2-side optical dies occupy only 0.43 mm^2 and 1.21 mm^2 respectively. Edge couplers add minimal area overhead along package boundaries. Compared to conventional NVLink interface (1.8 mm^2 [7]), our SM-side die achieves 76% area reduction while the L2-side die saves 33% area. The total optical die footprint per GPU is 2.93 mm^2, representing a modest 1.63× increase over NVLink - a reasonable tradeoff given the performance benefits.

2) Optical Switch: Our implementation leverages a Delta network topology for the centralized optical switch, co-packaged with the electronic die. This topology choice optimizes both cost and performance by minimizing MR count compared to alternative topologies (fat-tree or crossbar) while maintaining comparable blocking characteristics [28]. The optical dies integrate with GPU electronic dies through 3D stacking technology, following industry-proven approaches [13] [33]. While the 3D stacking approach increases manufacturing complexity and cost, such advanced packaging is justified in high-end markets such as data center and HPC.

TABLE IV: Optical device cost per GPU

Component	SM-side	L2-side	Total
Fiber count	10	42	52
Waveguides	36	216	252
Transceiver MR	5504	15456	21472
Optical switching MR	0	512	

VI. Conclusion

We have presented DOME, a coordinated control scheme for optical networks in rack-scale multi-GPU systems. Our design features an inter-node interface that bridges inter/intra-GPU network domains and implements the multi-switch and multi-domain control scheme. Evaluation results demonstrate a 64% speedup and 31% energy reduction compared to electrical switches, along with a 14% speedup and 3% energy reduction over state-of-the-art approaches. Future work envisions extending DOME to connect disaggregated memory architectures and software-guided scheduling optimizations for collective communication patterns.

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